

1.1 Basic Set Concepts



Figure 1.2 A spoon, fork, and knife are elements of the set of flatware. (credit: modification of work “Cupofjoy” Wikimedia CC0 1.0 Public Domain Dedication)

Learning Objectives

After completing this section, you should be able to:

1. Represent sets in a variety of ways.
2. Represent well-defined sets and the empty set with proper set notation.
3. Compute the cardinal value of a set.
4. Differentiate between finite and infinite sets.
5. Differentiate between equal and equivalent sets.

Sets and Ways to Represent Them

Think back to your kitchen organization. If the drawer is the **set**, then the forks and knives are **elements** in the set. Sets can be described in a number of different ways: by roster, by set-builder notation, by interval notation, by graphing on a number line, and by Venn diagrams. Sets are typically designated with capital letters. The simplest way to represent a set with only a few members is the **roster** (or *listing*) **method**, in which the elements in a set are listed, enclosed by curly braces and separated by commas. For example, if F represents our set of flatware, we can represent F by using the following set notation with the roster method:

$$F = \{\text{fork, spoon, knife, meat thermometer, can opener}\}$$

EXAMPLE 1.1

Writing a Set Using the Roster or Listing Method

Write a set consisting of your three favorite sports and label it with a capital S .

Solution

There are multiple possible answers depending on what your three favorite sports are, but any answer must list three different sports separated by commas, such as the following:

$$S = \{\text{hockey, basketball, soccer}\}$$

YOUR TURN 1.1

1. Write a set consisting of four small hand tools that might be in a toolbox and label it with a capital T .

All the sets we have considered so far have been **well-defined sets**. A well-defined set clearly communicates whether an element is a member of the set or not. The members of a well-defined set are fixed and do not change over time. Consider the following question. What are your top 10 songs of 2021? You could create a list of your top 10 favorite songs from 2021, but the list your friend creates will not necessarily contain the same 10 songs. So, the set of your top 10 songs of 2021 is not a well-defined set. On the other hand, the set of the letters in your name is a well-defined set because it does not vary (unless of course you change your name). The NFL wide receiver, Chad Johnson, famously

changed his name to Chad Ochocinco to match his jersey number of 85.

EXAMPLE 1.2

Identifying Well-Defined Sets

For each of the following collections, determine if it represents a well-defined set.

1. The group of all past vice presidents of the United States.
2. A group of old cats.

✓ Solution

1. The group of all past vice presidents of the United States is a well-defined set, because you can clearly identify if any individual was or was not a member of that group. For example, Britney Spears is not a member of this set, but Joe Biden is a member of this set.
2. A group of old cats is not a well-defined set because the word old is ambiguous. Some people might consider a seven-year-old cat to be old, while others might think a cat is not old until it is 13 years old. Because people can disagree on what is and what is not a member of this group, the set is not well-defined.

> YOUR TURN 1.2

For each of the following collections, determine if it represents a well-defined set.

1. A collection of medium-sized potatoes.
2. The original members of the Black Eyed Peas musical group.

On January 20, 2021, Kamala Harris was sworn in as the first woman vice president of the United States of America. If we were to consider the set of all women vice presidents of the United States of America prior to January 20, 2021, this set would be known as an **empty set**; the number of people in this set is 0, since there were no women vice presidents before Harris. The empty set, also called the null set, is written symbolically using a pair of braces, $\{ \}$, or a zero with a slash through it, $\emptyset$.

 The set containing the number 0, $\{0\}$, is a set with one element in it. It is not the same as the empty set, $\{ \}$, which does not have any elements in it. Symbolically: $\{0\} \neq \{ \}$.

EXAMPLE 1.3

Representing the Empty Set Symbolically

Represent each of the following sets symbolically.

1. The set of prime numbers less than 2.
2. The set of birds that are also mammals.

✓ Solution

1. A prime number is a natural number greater than 1 that is only divisible by one and itself. Since there are no prime numbers less than 2, this set is empty, and we can represent it symbolically as follows: $\{ \}$ or $\emptyset$. These two different symbols for the empty set can be used interchangeably.
2. The set of birds and the set of mammals do not intersect, so the set of birds that are also mammals is empty, and we can represent it symbolically as $\emptyset$ or $\{ \}$.

> YOUR TURN 1.3

1. Represent the set of all numbers divisible by 0 symbolically.

?? WHO KNEW?

The Number Zero

We use the number zero to represent the concept of nothing every day. The machine language of computers is binary, consisting only of zeros and ones, and even way before that, the number zero was a powerful invention that allowed our understanding of mathematics and science to develop. The historical record shows the Babylonians first used zeros around 300 B.C., while the Mayans developed and began using zero separately around 350 A.D. What is considered the first formal use of zero in arithmetic operations was developed by the Indian mathematician Brahmagupta around 650 A.D.

[Brahmagupta, Mathematician and Astronomer \(https://openstax.org/r/brahmagupta.html\)](https://openstax.org/r/brahmagupta.html)

Another interesting feature of the number zero is that although it is an even number, it is the only number that is neither negative nor positive.

For larger sets that have a natural ordering, sometimes an ellipsis is used to indicate that the pattern continues. It is common practice to list the first three elements of a set to establish a pattern, write the ellipsis, and then provide the last element. Consider the set of all lowercase letters of the English alphabet, A . This set can be written symbolically as $A = \{a, b, c, \dots, z\}$.

The sets we have been discussing so far are **finite sets**. They all have a limited or fixed number of elements. We also use an ellipsis for **infinite sets**, which have an unlimited number of elements, to indicate that the pattern continues. For example, in set theory, the set of **natural numbers**, which is the set of all positive counting numbers, is represented as $\mathbb{N} = \{1, 2, 3, \dots\}$.

Notice that for this set, there is no element following the ellipsis. This is because there is no largest natural number; you can always add one more to get to the next natural number. Because the set of natural numbers grows without bound, it is an infinite set.

EXAMPLE 1.4

Writing a Finite Set Using the Roster Method and an Ellipsis

Write the set of even natural numbers including and between 2 and 100, and label it with a capital E . Include an ellipsis.

✓ Solution

Write the label, E , followed by an equal sign and then a bracket. Write the first three even numbers separated by commas, beginning with the number two to establish a pattern. Next, write the ellipsis followed by a comma and the last number in the list, 100. Finally, write the closing bracket to complete the set.

Write the label, E , followed by an equal sign and then a bracket.

$$E = \{$$

Write the first three even numbers separated by commas, beginning with the number 2 to establish a pattern.

$$E = \{2, 4, 6,$$

Next, write the ellipsis followed by a comma and the last number in the list, 100.

$$E = \{2, 4, 6, \dots, 100$$

Finally, write the closing bracket to complete the set.

$$E = \{2, 4, 6, \dots, 100\}$$

> YOUR TURN 1.4

1. Use an ellipsis to write the set of single digit numbers greater than or equal to zero and label it with a capital D .

Our number system is made up of several different infinite sets of numbers. The set of **integers**, $\mathbb{Z}$, is another infinite set of numbers. It includes all the positive and negative counting numbers and the number zero. There is no largest or smallest integer.

EXAMPLE 1.5**Writing an Infinite Set Using the Roster Method and Ellipses**

Write the set of integers using the roster method, and label it with a $\mathbb{Z}$.

✓ Solution

Step 1: As always, we write the label and then the opening bracket. Because the negative counting numbers are infinite, to represent that the pattern continues without bound to the left, we must use an ellipsis as the first element in our list.

Step 2: We place a comma and follow it with at least three consecutive integers separated by commas to establish a pattern.

Step 3: Add an ellipsis to the end of the list to show that the set of integers continues without bound to the right.

Complete the list with a closing bracket. The set of integers may be represented as follows:

$$\mathbb{Z} = \{ \dots, -2, -1, 0, 1, 2, \dots \}.$$

> YOUR TURN 1.5

1. Write the set of odd numbers greater than 0 and label it with a capital M .

A shorthand way to write sets is with the use of **set builder notation**, which is a verbal description or formula for the set. For example, the set of all lowercase letters of the English alphabet, A , written in set builder notation is:

$$A = \{x | x \text{ is a lowercase letter of the English alphabet.}\}$$

This is read as, "Set A is the set of all elements x such that x is a lowercase letter of the English alphabet."

EXAMPLE 1.6**Writing a Set Using Set Builder Notation**

Using set builder notation, write the set B of all types of balls. Explain what the notation means.

✓ Solution

The verbal description of the set is, "Set B is the set of all elements b such that b is a ball." This set can be written in set builder notation as follows: $B = \{b | b \text{ is a ball.}\}$

> YOUR TURN 1.6

1. Using set builder notation, write the set C of all types of cars.

EXAMPLE 1.7**Writing Sets Using Various Methods**

Consider the set of letters in the word "happy." Determine the best way to represent this set, and then write the set using

either the roster method or set builder notation, whichever is more appropriate.

Solution

Because the letters in the word “happy” consist of a small finite set, the best way to represent this set is with the roster method. Choose a label to represent the set, such as H .

$$H = \{h, a, p, y\}.$$

Notice that the letter “p” is only represented one time. This occurs because when representing members of a set, each unique element is only listed once no matter how many times it occurs. Duplicate elements are never repeated when representing members of a set.

YOUR TURN 1.7

1. Use the roster method or set builder notation to represent the collection of all musical instruments.

Computing the Cardinal Value of a Set

Almost all the sets most people work with outside of pure mathematics are finite sets. For these sets, the **cardinal value** or **cardinality** of the set is the number of elements in the set. For finite set A , the cardinality is denoted symbolically as $n(A)$. For example, a set that contains four elements has a cardinality of 4.

How do we measure the cardinality of infinite sets? The ‘smallest’ infinite set is the set of natural numbers, or counting numbers, $\mathbb{N} = \{1, 2, 3, \dots\}$. This set has a cardinality of $\aleph_0$ (pronounced “aleph-null”). All sets that have the same cardinality as the set of natural numbers are **countably infinite**. This concept, as well as notation using aleph, was introduced by mathematician Georg Cantor who once said, “A set is a Many that allows itself to be thought of as a One.”

EXAMPLE 1.8

Computing the Cardinal Value of a Set

Write the cardinal value of each of the following sets in symbolic form.

1. $F = \{\text{fork, spoon, knife, meat thermometer, can opener}\}$
2. The empty set.

Solution

1. There are 5 distinct elements in set F : a fork, a spoon, a knife, a meat thermometer, and a can opener. Therefore, the cardinal value of set F is 5 and written symbolically as $n(F) = 5$.
2. Because the empty set does not have any elements in it, the cardinality of the empty set is zero. Symbolically we write this as: $n(\emptyset) = 0$.

YOUR TURN 1.8

Write the cardinal value of each of the following sets in symbolic form.

1. Set P is the set of prime numbers less than 2.
2. Set A is the set of lowercase letters of the English alphabet, $A = \{a, b, c, \dots, z\}$.

Now that we have learned to represent finite and infinite sets using both the roster method and set builder notation, we should also be able to determine if a set is finite or infinite based on its verbal or symbolic description. One way to determine if a set is finite or not is to determine the cardinality of the set. If the cardinality of a set is a natural number, then the set is finite.

EXAMPLE 1.9**Differentiating Between Finite and Infinite Sets**

Classify each of the following sets as infinite or finite.

1. $E = \{2, 4, 6, 8, 10\}$
2. A is the set of lowercase letters of the English Alphabet, $A = \{a, b, c, \dots, z\}$.
3. $\mathbb{Q} = \left\{ \frac{p}{q} \mid p \text{ and } q \text{ are integers and } q \neq 0 \right\}$

✓ **Solution**

1. $n(E) = 5$. Since 5 is a natural number, the set is finite.
2. $n(A) = 26$. Since 26 is a natural number, the set is finite.
3. Set $\mathbb{Q}$ is the set of rational numbers or fractions. Because the set of integers is a subset of the set of rational numbers, and the set of integers is infinite, the set of rational numbers is also infinite. There is no smallest or largest rational number.

> **YOUR TURN 1.9**

Classify each of the following sets as infinite or finite.

1. $B = \{b, a, k, e\}$
2. $\mathbb{R} = \{x \mid x \text{ is a real number}\}$

Equal versus Equivalent Sets

When speaking or writing we tend to use equal and equivalent interchangeably, but there is an important distinction between their meanings. Consider a new Ford Escape Hybrid and a new Toyota Rav4 Hybrid. Both cars are hybrid electric sport utility vehicles; in that sense, they are **equivalent**. They will both get you from place to place in a relatively fuel-efficient way. In this example we are comparing the single member set {Toyota Rav4 Hybrid} to the single member set {Ford Escape Hybrid}. Since these two sets have the same number of elements, they are also equivalent mathematically, meaning they have the same cardinality. But they are not equal, because the two cars have different looks and features, and probably even handle differently. Each manufacturer will emphasize the features unique to their vehicle to persuade you to buy it; if the SUVs were truly equal, there would be no reason to choose one over the other.

Now consider two Honda CR-Vs that are made with exactly the same parts, on the same assembly line within a few minutes of each other—these SUVs are **equal**. They are identical to each other, containing the same elements without regard to order, and the only differentiator when making a purchasing decision would be varied pricing at different dealerships. The set {Honda CR-V} is equal to the set {Honda CR-V}. Symbolically, we represent equal sets as $A = B$ and equivalent sets as $A \sim B$.

Now, let us consider a Toyota dealership that has 10 RAV4s on the lot, 8 Prii, 7 Highlanders, and 12 Camrys. There is a one-to-one relationship between the set of vehicles on the lot and the set consisting of the number of each type of vehicle on the lot. Therefore, these two sets are equivalent, but not equal. The set {RAV4, Prius, Highlander, Camry} is equivalent to the set {10, 8, 7, 12} because they have the same number of elements.

▶ **VIDEO**

[Equal and Equivalent Sets \(https://openstax.org/r/Equal_and_Equivalent_Sets\)](https://openstax.org/r/Equal_and_Equivalent_Sets)

⚠ If two sets are equal, they are also equivalent, because equal sets also have the same cardinality.

EXAMPLE 1.10**Differentiating Between Equivalent and Equal Sets**

Determine if the following pairs of sets are equal, equivalent, or neither.

1. $E = \{2, 4, 6, 8, 10\}$ and $F = \{\text{fork, spoon, knife, meat thermometer, can opener}\}$
2. The empty set and the set of prime numbers less than 2.
3. The set of vowels in the word happiness and the set of consonants in the word happiness.

✓ **Solution**

1. Sets E and F both have a cardinal value of 5, but the elements in these sets are different. So, the two sets are equivalent, but they are not equal: $E \sim F$.
2. The set of prime numbers consists of the set of counting numbers greater than one that can only be divided evenly by one and itself. The set of prime numbers less than 2 is an empty set, since there are no prime numbers less than 2. Therefore, these two sets are equal (and equivalent).
3. The set of vowels in the word happiness is $\{a, i, e\}$ and the set of consonants in the word happiness is $\{h, p, n, s\}$. The cardinal value of these sets two sets is $n(\{a, i, e\}) = 3$ and $n(\{h, p, n, s\}) = 4$, respectively. Because the cardinality of the two sets differs, they are not equivalent. Further, their elements are not identical, so they are also not equal.

> **YOUR TURN 1.10**

Determine if the following pairs of sets are equal, equivalent, or neither.

1. Set $B = \{b, a, k, e\}$ and set $A = \{a, b, e, k\}$
2. Set $B = \{b, a, k, e\}$ and set $F = \{f, l, a, k, e\}$
3. Set $B = \{b, a, k, e\}$ and set $C = \{c, a, k, e\}$



PEOPLE IN MATHEMATICS

Georg Cantor

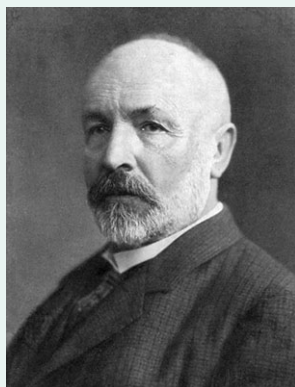


Figure 1.3 Georg Cantor (credit: Wikimedia, public domain)

Georg Cantor, the father of modern set theory, was born during the year 1845 in Saint Petersburg, Russia and later moved to Germany as a youth. Besides being an accomplished mathematician, he also played the violin. Cantor received his doctoral degree in Mathematics at the age of 22.

In 1870, at the age of 25 he established the uniqueness theorem for trigonometric series. His most significant work happened between 1874 and 1884, when he established the existence of transcendental numbers (also called irrational numbers) and proved that the set of real numbers are uncountably infinite—despite the objections of his former professor Leopold Kronecker.

Cantor published his final treatise on set theory in 1897 at the age of 52, and was awarded the Sylvester Medal from the Royal Society of London in 1904 for his contributions to the field. At the heart of Cantor's work was his goal to solve the continuum problem, which later influenced the works of David Hilbert and Ernst Zermelo.

References:

Wikipedia contributors. "Cantor." Wikipedia, The Free Encyclopedia, 23 Mar. 2021. Web. 20 Jul. 2021.

Akihiro Kanamori, "Set Theory from Cantor to Cohen," Editor(s): Dov M. Gabbay, Akihiro Kanamori, John Woods, *Handbook of the History of Logic*, North-Holland, Volume 6, 2012.

Check Your Understanding

1. A _____ is a well-defined collection of objects.
2. The _____ of a finite set A , denoted $n(A)$, is the number of elements in set A .
3. Determine if the following description describes a well-defined set: "The top 5 pizza restaurants in Chicago."
4. The United States is the only country to have landed people on the moon as of March 21, 2021. What is the cardinality of the set of all people who have walked on the moon prior to this date?
5. Set A is a set of a dozen distinct donuts, and set B is a set of a dozen different types of apples. Is set A equal to set B , equivalent to set B , or neither?
6. Is the set of all butterflies in the world a finite set or an infinite set?
7. Represent the set of all upper-case letters of the English alphabet using both the roster method and set builder notation.



SECTION 1.1 EXERCISES

For the following exercises, represent each set using the roster method.

1. The set of primary colors: red, yellow, and blue.
2. A set of the following flowers: rose, tulip, marigold, iris, and lily.
3. The set of natural numbers between 50 and 100.
4. The set of natural numbers greater than 17.
5. The set of different pieces in a game of chess.
6. The set of natural numbers less than 21.

For the following exercises, represent each set using set builder notation.

7. The set of all types of lizards.
8. The set of all stars in the universe.
9. The set of all integer multiples of 3 that are greater than zero.
10. The set of all integer multiples of 4 that are greater than zero.
11. The set of all plants that are edible.
12. The set of all even numbers.

For the following exercises, represent each set using the method of your choice.

13. The set of all squares that are also circles.
14. The set of natural numbers divisible by zero.
15. The set of Mike and Carol's children on the TV show, *The Brady Bunch*.
16. The set of all real numbers.
17. The set of polar bears that live in Antarctica.
18. The set of songs written by Prince.
19. The set of children's books written and illustrated by Mo Willems.
20. The set of seven colors commonly listed in a rainbow.

For the following exercises, determine if the collection of objects represents a well-defined set or not.

21. The names of all the characters in the book, *The Fault in Our Stars* by John Green.
22. The five greatest soccer players of all time.
23. A group of old dogs that are able to learn new tricks.
24. A list of all the movies directed by Spike Lee as of 2021.
25. The group of all zebras that can fly an airplane.
26. The group of National Baseball League Hall of Fame members who have hit over 700 career home runs.

For the following exercises, compute the cardinal value of each set.

27. $P = \{\text{Snuzzle, Butterscotch, Blue Belle, Minty, Blossom, Cotton Candy}\}$

28. $T = \{\text{pepperoni, sausage, bacon, ham, mushrooms, olives, bell pepper, pineapple}\}$
29. $\emptyset$
30. $B = \{5, 6, 7, \dots, 20\}$
31. $F = \{\frac{1}{9}, \frac{2}{9}, \frac{3}{9}, \frac{4}{9}, \frac{5}{9}, \frac{6}{9}, \frac{7}{9}, \frac{8}{9}, \frac{9}{9}\}$
32. $\{ \}$
33. $C = \{n^3 | n \text{ is a member of } \mathbb{N}\}$
34. $S = \{7n | n \text{ is an element of } \mathbb{N}\}$
35. $L = \{l, m, n, \dots, y\}$
36. The set of numbers on a standard 6-sided die.

For the following exercises, determine whether set A and set B are equal, equivalent or neither.

37. $A = \{\text{right, acute, obtuse}\}; B = \{\text{equilateral, scalene, isosceles}\}.$
38. $A = \{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}\}; B = \{\frac{1}{4}, \frac{1}{3}, \frac{1}{2}, 1\}.$
39. $A = \{\text{red, orange, yellow}\}; B = \{\text{green, blue, indigo, violet}\}.$
40. $A = \{5n | n \in \mathbb{N}\}; B = \mathbb{N}.$
41. $A = \{-2, -1, 0, \dots\}; B = \{2, 3, 5, \dots\}.$
42. $A = \{\text{John, Paul, George, Ringo}\}; B = \{\text{Bono, Larry, The Edge, Adam}\}.$
43. $A = \emptyset; B = \{ \}.$
44. $A = \{\text{lemon, lime, orange}\}; B = \{\text{orange, lemon, lime, grape}\}.$

For the following exercises, determine if the set described is finite or infinite.

45. The set of natural numbers.
46. The empty set.
47. The set consisting of all jazz venues in New Orleans, Louisiana.
48. The set of all real numbers.
49. The set of all different types of cheeses.
50. The set of all words in *Merriam-Webster's Collegiate Dictionary*, Eleventh Edition, published in 2020.

1.2 Subsets



Figure 1.4 The players on a soccer team who are actively participating in a game are a subset of the greater set of team members. (Credit: "PAFC-Mezokovesd-108" by Puskás Akadémia/Flickr, Public Domain Mark 1.0)

Learning Objectives

After completing this section, you should be able to:

1. Represent subsets and proper subsets symbolically.
2. Compute the number of subsets of a set.
3. Apply concepts of subsets and equivalent sets to finite and infinite sets.

The rules of Major League Soccer (MLS) allow each team to have up to 30 players on their team. However, only 18 of these players can be listed on the game day roster, and of the 18 listed, 11 players must be selected to start the game.